



A study of changes in education in Aarhus

EDUCATION-LEVEL IN THE 1800'S

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Portfolios	2
Final Project – Education-level in Aarhus, in the period 1801-1845.....	15
Introduction for education in the period 1801-1845.	15
Background information for the period in interest	15
Methods	17
Finding articles	19
Our Findings	20
Graph over profession from 1801.....	20
Graph over profession from 1834.....	21
Graph over profession from 1845.....	22
Profession of women.	22
Graph over work for craftsmen, men and women.	23
Graph over "elev" (student).	24
Graph over "lærlinge"	24
Graph over teachers.	25
Discussion.....	26
School in Aarhus.	30
Women in school.	31
Lessons learned	33
Conclusion	33
Author contributions	35
Table 1 – Software metadata.....	35
Table 2 – Data metadata (use the template below or create your own metadata table)	36
Bibliography.....	37
Deklaration for anvendt Generativ Kunstig Intelligens (GAI)	41

Portfolios

Prtf1 Regex and OpenRefine

Task 1: List regex that extracts dates from <http://bit.ly/regexexercise2> and formats them as YYYY-MM-DD?

URL for the solution: <https://regex101.com/r/ucL0Uf/1>

Brief explanation: We started with copying the text from “Regex CD Public discourse” in to the program. An then we made 3 groups of parentheses (). Then we typed in \d into group 1 to collect digits and to then range them from 1 to 2 digtes {1,2}, to collect the numbers of days. Then we did the same again to collect the months in group 2. Then we collected the years by typing \s to match any whitespace character. Then we added a ? to match the previous tokens between 0 and 1, as many times as needed. Then for the third and final group which was like the first two with \d{4} to match the previous token exactly 4 times. Then to the substitute function to write in \$3-\$2-\$1 so that the groups was in the right order.

Task 2 Regex to convert stopwordlist between R and Voyant format

R to Voyant: <https://regex101.com/r/LpuxOe/1>

Brief explanation: We began with copying in the stop word list from the given document. Then we wrote “, “ into regular expression search-bar to catch alle the quotation marks and the white spaces in-between the words. Then we wrote in \n into the substitution search bar to substitute the quotation marks and the white spaces with line break

Voyant to R: <https://regex101.com/r/BH83ob/1>

Brief explanation: We began with copying in the stop word list from the given document. Then we wrote \n into regular expression search-bar to catch line breaks in-between the

words. Then we wrote in “, “ into the substitution search bar to substitute line breaks with quotation marks and a white space

Task 3 Does OpenRefine alter the raw data during sorting and filtering?

YES / NO, it does not

Task 4: What are the two most water-deprived months in Mozambique and Tanzania?

We started with putting in the given data into open refine. Then we reorganized the columns so that interview dates and the “no_enough-water” column was beside one another. Then we transformed the interview dates into dates and opened up a timeline facet for the interview dates and a text facet for the the “no_enough-water” column. Then vi included all the no answers in the “no_enough-water” column so we only could see the dates where there was no water

Answer: November and May

Task 5: Challenge - 10 most common occupations among men in 1801 Aarhus

Total number of people in your selected cohort: 543

Define what is / is not an occupation:

An occupation is a job in which you get paid in some kind of way (usually in either money or food and lodging)

10 most frequent occupations (list number of people practising that job with each entry)

Soldat: 367 men

- Tjenestekarl: 37 men
- Rytter: 32 men
- Væver: 30 men
- Matros: 17 men

- Recrut: 17 men
- Skriverkarl: 12 men
- Gårdskarl: 11 men
- Bonde: 10 men
- Daglejer: 10 men

Prtf 2 Danish Monarchs

Github URL for your rmarkdown/html files: <https://github.com/Digital-Methods-HASS>

.....

Answer these questions by listing both the chunk of code you used and then the response

What function did you use to load your dataset in R as a tibble? What do you see when you run head() on your 'kings' digital object?

How do you confirm your kings' object in R is a tibble?

We used this function: `kings <- read_csv("data/kings.csv", na = c("NULL", ""))`. Because we use an underscore in read_csv the dataset loads as a tibble.

When we run the head(kings) function we get a view of the first 6 kings/rows in the dataset.

A tibble: 6 x 12

Name	House	Start_year	End_year	Birth_year	Death_year	Gender	Dynasty
Gorm den Gamle	Gorm	936	958	908	958	M	Jellingdynastiet
Harald 1. Blåtand	Gorm	NA	NA	936	985	M	Jellingdynastiet
Toke, Gormsen	Gorm	985	986	NA	986	M	Jellingdynastiet
Svend 1. Tveskæg	Gorm	NA	NA	NA	1014	M	Jellingdynastiet
Harald 2.	Gorm	1014	1018	963	NA	M	Jellingdynastiet
Knud 1. den Store	Gorm	1018	1035	NA	1035	M	Jellingdynastiet

Paste here a screenshot of the Console after running **head(kings)** or **glimpse(kings)**:

Here is a screenshot of the Console after running **glimpse(kings)**

```
> glimpse(kings)
Rows: 57
Columns: 12
 $ Name      <chr> "Gorm den Gamle", "Harald 1. Blåtand", "Toke, Gormsen", "Svend 1. Tveskæg", "Harald 2.", "Knud 1. den Stor-
 $ House     <chr> "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Gorm", "Fairhair", "Estridsen", "Estridsen",
 $ Start_year <dbl> 936, NA, 985, NA, 1014, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1104, 1134, 1137, 1146, 1157, 118
 $ End_year   <dbl> 958, NA, 986, NA, 1018, 1035, 1042, 1047, 1074, 1080, 1086, 1095, 1103, 1134, 1137, 1146, 1157, 1182, 120
 $ Birth_year <dbl> 908, 936, NA, NA, 963, NA, 995, 1018, 1024, 1039, 1041, 1043, 1050, 1055, 1065, 1100, 1120, 1131, 1163, 1
 $ Death_year <dbl> 958, 985, 986, 1014, NA, 1035, 1042, 1047, 1076, 1080, 1086, 1095, 1103, 1134, 1137, 1146, NA, 1182, 1202
 $ Gender     <chr> "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M", "M",
 $ Dynasty    <chr> "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jellingdynastiet", "Jell
 $ Source     <chr> "https://da.wikipedia.org/wiki/Gorm_den_Gamle", "https://da.wikipedia.org/wiki/Harald_2./Knud_1./Bastard", "Net
 $ Birthdmy   <chr> "908", "936", NA, "17/04/963", "994", "995", "1018", "1024", "1019", "1041", "1043", "1050", "1055", "113
 $ Deathdmy   <chr> "958", "987", NA, "03/02/1014", "1018", "12/11/1035", "08/06/1042", "25/10/1047", "28/04/1076", "17/04/10
 $ Duration   <dbl> 22, NA, 1, NA, 4, 17, 7, 5, 27, 6, 6, 9, 8, 30, 3, 9, 11, 25, 20, 39, 9, 2, 7, 27, 33, 7, 3, 3, 35, 12, 9.
```

1. What is the average age of reign among your monarchs and how many rulers have exceeded it?

The Code

```

#1 Finds the kings who enjoyed a longer-than-average duration of reign
long_reigns <- kings %>%
  filter(Duration > avg_reign_vector)
long_reigns

#2 Shows which kings enjoyed a longer-than-average duration of reign
long_reigns %>%
  select(Name, Duration) %>%
  arrange(desc(Duration))

# 5. Counts how many kings enjoyed a longer-than-average duration of reign
long_reigns %>%
  count()

```

Result/Answer

The average age of reign among danish monarchs is 19,5769 $\approx$ 19,6 years and 25 kings enjoyed a longer-than-average duration of reign.

How many days did the three longest-ruling monarchs rule?

The Code

```

'''(r)
# YOUR CODE

top_3_monarchs <- kings %>%
  arrange(desc(Duration)) %>%
  slice(1:3) %>%
  mutate(
    Days_Basic = Duration * 365,
    # To count transition/leap year we add 0.25 to the 365 days, because every fourth year is a leap year
    Days_Leap = Duration * 365.25
  )

print(top_3_monarchs %>% select(Name, Duration, Days_Leap))
'''

# A tibble: 3 x 3
  Name          Duration Days_Leap
  <chr>          <dbl>    <dbl>
1 Christian 4.    60      21915
2 Margrete 2.    51      18628.
3 Erik 7. af Pommern 43      15706.

```

Result/Answer

The three longest-ruling monarchs were Christian 4., Magrethe 2. and Erik 7. of Pommern. Christian 4. ruled for 21915 days. Magrethe 2. ruled for 18628 days. Erik 7. of Pommern ruled for 15706 days.

A little fun-fact is that, Christian.4 became king when he was just 11 years old, after his farther Federik.2 died, and he became under guardianship. He was the king, but not the ruler.

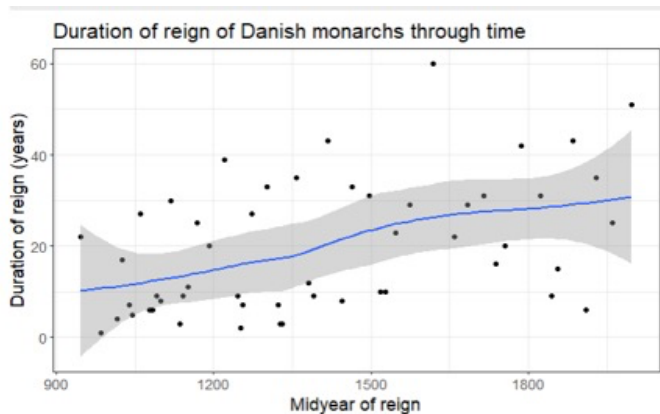
Challenge: Plot the duration of reign in time and briefly answer “What is the long-term trend in the duration of reign among monarchs?”

The Code

```
kings <- kings |>
  mutate(Midyear = Start_year + (End_year - Start_year) / 2)

ggplot(kings, aes(x = Midyear, y = Duration)) +
  geom_point() +
  geom_smooth() +
  labs(
    title = "Duration of reign of Danish monarchs through time",
    x = "Midyear of reign",
    y = "Duration of reign (years)"
  ) +
  theme_bw() +
  theme(text = element_text(size = 14))
```

Visual



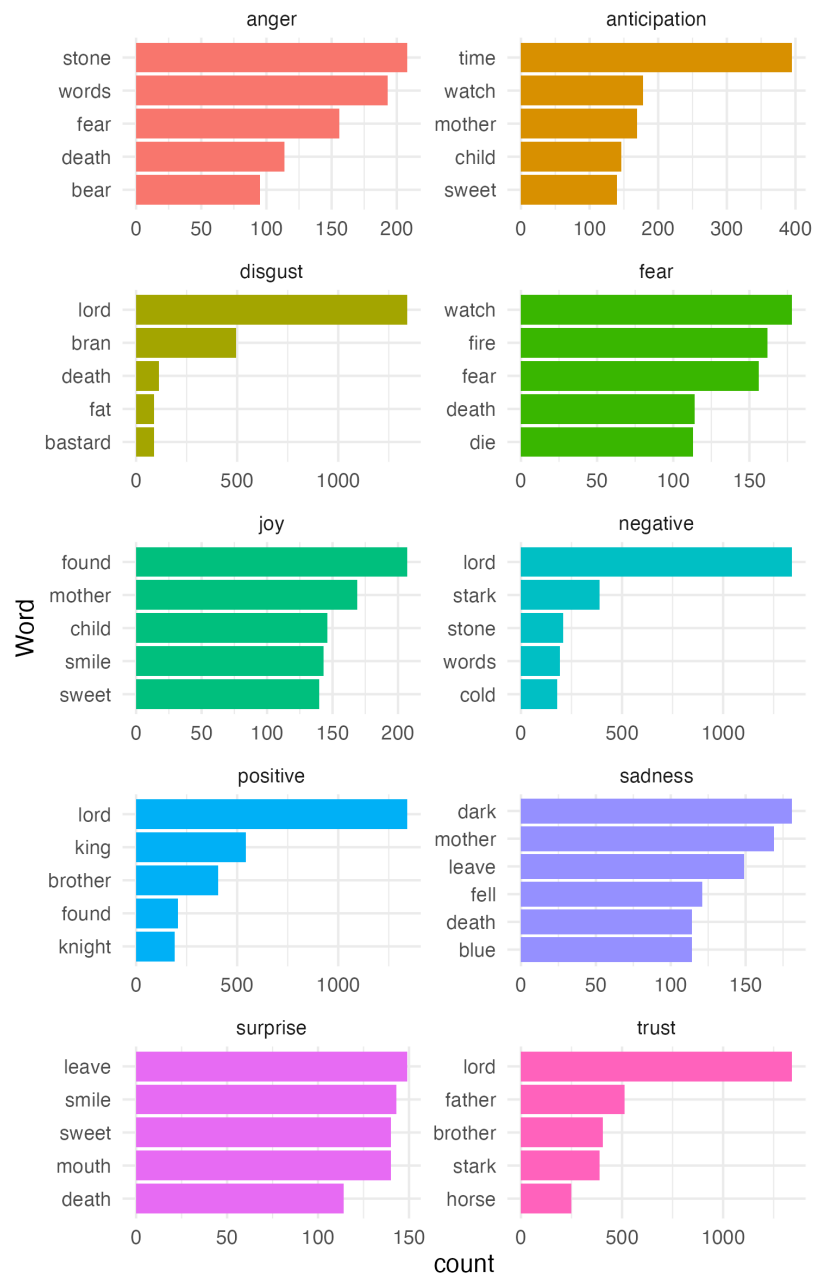
Answer

The visual shows that there is a long-term trend of an increasing duration of reign among the Danish monarchs. The blue line visualizes the average.

Prtf 3 [Game of thrones sentiment analysis]

URL to Github code: https://github.com/Digital-Methods-HASS/AU791343_Krogh_Kristian

Visualisation:



Significance: Explain what your visualisation represents and what are its implications vis-a-vis your [historical] research question (250 words)? Also, consider briefly what your visual does not show.

My visualisation displays what kind of words/language that George R. Martin used in his writing of his book. It also shows the quantity of these words in the text.

By looking at this visualisation, we can see that the most used meaningful words were Stone, time, lord, watch, found, dark and leave. We can also see that according to R, George R. Martin most commonly used words that had the connotations of: Disgust, Negative, Positive and Trust. This follows the expatiations that I had before running this analysis. Because, I had read the book beforehand (and watch the tv-series), I knew that the themes of the book were fantasy, knighthood and betrayal which all is within the emotion R tells me, is used the most.

But that said. The visualisation also shows me that R has understood the word “lord” in four different ways (namely as having the connotations: Disgust, Negative, Positive and Trust). This speaks to that the “nrc”-library has listed the word “lord” under four different categories. This is a problem because it makes the sentiment of text becomes ambiguous because the visualisation does not show in which contexts these words appear in the text. So the visualisation does not give us a clear answer to the question of which sentiment that George R Martin wanted to express in his book and by not doing that, that the visualisation and our analysis in R does not replace close reading but just supplements it and shows us were close reading is needed

Prtf 4 [Digital archives reflection assignment]

URL to Github code:

https://github.com/KwolfKrogh/DAM_Final_project_2026/tree/056cfe8ed81c0e2eeadd eb1635fb12001b691e69/Uddannelse_i%3AAarhus_1800/Portfolie_opgaver

1. Explain how Jensen's (2021) category of "digital archival literacy" addresses the distinct challenges digital archives represent for historical research. How has digitization changed historians' interactions with their sources? In your response, focus on two points that you found most thought-provoking. You may also choose to cite Jensen and Schriver (2022) in your response.

In her article from 2020¹, Jensen argues that, because sources are being digitized, historians are beginning to face new kinds of problems and opportunities. She, for example, argues that the way that the digital archives are displayed and organized influences the way one that you view sources and which sources that you find. For example, if you wanted to research a topic, before the digital archives, you would go down to the library and find the information in traditional sources. While you were doing that you would encounter metadata. With other words you get a bunch of “data” that is related to the sources witch you then must distinguish between. That is not the case with digital archives. When you look up sources you don’t automatically (at least in the same way) get a bunch of related data that you as an historian must make up one's mind about. For example, the archive can be founded by an organization or project that have particular interest or agendas (for example that it should speak to a broad audience or a certain topic). This influence which sources that are chosen to be preserved and digitized. And thus, influencing what kind of data that we as historians are presented. Jensen there for argues that it is important as historians to think about what data therefor was omitted² and why it was omitted, since these kinds of meta data often can lead to perspectives on the subject or play a role in the creditability of the digital source.

2. Briefly provide the biographical details of the interviewee.

¹ Helle Jensen, “Helle Strandgaard Jensen”, i *Media History* (2020, 2020), <https://doi.org/10.1080/13688804.2020.1779047>.

² Jensen, “Helle Strandgaard Jensen”, 6.

I choose the interview with Baruch Friedman³, done by Davide Boder the 2 of September 1946

The one being interviewed is Baruch Friedman, a Slovakian Zionist that did not believe in the official story about Jews being relocated into workcamps and therefor hid in a forest with his younger brother with the help of a gentile acquaintance. Eventually when the deportation had stopped the brothers reported voluntarily to one of the Slovekian workcamps (which was relatively agreeable). Freedmen and his younger brother were freed along with other Jews when the uprising against Tiso's regime happened in august 28 - October 2, 1944. After having been freed both Friedman fought in the uprising (but Friedman did have his resevations with both the anti-nazi Slovaks and the uprisings chances for success). At the time that Friedman was interviewed he was living with his bother in Italy.

3. After reading the "description" panel, discuss the provenance of the source. For example, what metadata is provided by the archive, and which historical details did the archive choose to emphasize in their written summary of the recording?

The first thing that I noticed when I was analyzing the provenance of Baruch Friedman story was that the Illinois institute of technology/David Boder had chosen to record and tag which subjects the recording fitted under. Which was, for example: "World War, 1939-1945--Refugees". This is interesting because it tells us what aspects of the interviews that Boder used to categories and order the interviews in and there what aspects of the interview that he found important. The same can be said with the beforehand described summery of the interview. He we can see that Boder thought that it was important to record how and why Friedman ended up in a work camp and what happened with him and his bother after that they had been freed. It is also evident that Boder found it relevant (and jet another way of organizing/categorizing the interviews) to record what city Friedmann had come from, since "Kurima (Slovakia)" also is a tag given to the interview. Next, Boder/ the Illinois institute of technology chose to record the different people that had a relation to the interview (for example, that the interviewee was Baruch Friedmann and the interviewer: David P. Boder). This is interesting because it shows an interest in the interview's different participants (and

³ Baruch Friedman, "David P. Boder Interviews Baruch Friedman", (Tradate Italien), Illinois institute of Technology, 2. september 1946, Lydoptagelse, Voices of The Holocaust, [https://iit.aviaryplatform.com/collections/231/collection_resources/17610?u=t&keywords\[\]=Baruch&keyword s\[\]=Friedman](https://iit.aviaryplatform.com/collections/231/collection_resources/17610?u=t&keywords[]=Baruch&keyword s[]=Friedman).

later handlers), which gives us an idea of the amount of handling/interpretation (and therefore with its level of credibility) the interview has gone through before being archived. Lastly, the archive chose to include that Friedman's story have the source-types of "interview" and "oral history".

4. What are the implications of using interview recordings as a historical source?

The implications of interview recording as a historical source is that we as listeners must interpret what is being communicated. This does maybe not sound as much but as opposed to written historical sources (where the writer writes his intentions/feelings plainly) the sound recordings of interviews requires that the listeners to interpret themselves what are being said and how it is being said (is the interviewee sad, happy or something third?). This means that an element of ambiguity gets introduced which we, as historians, not always are interested in since it leads to confusion and strife since to different people can have slightly different interpretation of the recorded interview. That said, it is also a benefit of recording interviews as historical sources that we get a feeling of the situation that the interview was conducted, which we could not get by reading a traditional source. One of the other implications interview recordings have been that an interview is limited in the information it gives by the questions that are asked by the interviewer. Therefore the interview (and therefore the source itself) is formed by the desires, curiosities and ideas of importance the interviewer may have and not what the interviewee could have found important. Lastly, it is also worth mentioning that the interview recording themselves does not provide any tags or keywords which means that they are difficult to find when searching the digital archives, which then limits the availability of the source. That also means that these tags and keywords, must be provided by translators/transcribers which themselves interpret the source and by doing so influences the general understanding of source.

5. Compare and contrast the use of digitized historical sources in web-based databases with reborn digital sources in the archived Web. Should these sources be treated differently when conducting historical research? To complete this task, draw on your knowledge of at least two database collections discussed in class and/or your readings. In your response, please support your position using relevant literature from our course, such as Kelley (2019), Johnson (2018), Brügger (2019), and the articles mentioned in part one of this assignment.

I think that both types of historical sources are useful for us as historians. This I think because it is my experience that digitized historical sources (for example the documents that the royal Danish library has been/are digitizing, as pictures on their website) and the reborn digital sources with (for example the “wayback machine” which we were introduced in class) can tell us different things about a chosen topic. That said, these kinds of digital sources are not in any way infallible.

First, it is important, when working with digital sources to consider how the different sources were made. As Brügger⁴ mentions in her article from 2019, the reborn digital sources are made by scanning a given website, with no certainty that when the website did not change while or after the program had finished copying the website. This is important to consider because it makes it, so the reborn digital sources are made of fragments. This means that the reborn websites that find in the archived web is made up of different pieces that were not necessarily made (and thus not displayed) at the same time. This is important to consider if your goal is to use the reborn digital archive to illuminate a specific time/date, because you can't be sure that the portrayal of the web site that you are presented by the web archive are the exactly the same as you would have met if you had visited the website within the relevant period.

This is a problem that the digitized historical sources do not have. Because the digitized historical source of is a picture or a screenshot that a program or person at some times have taken (and thus not an attempt to copy of the source) the content of the archived material does not have the risk of having been stitched together using chronological-different pieces.

The real difference between these two kinds of sources is the amount and way we as research can interact with the past. What is meant by this is that, when you access a former version of a chosen website through a web archive, you get to interact with the website as if you used it while it was the current version. You get to click on the website's different hyperlinks, to see what kind of material/topics that the web site made references to and so on (although it should be said that because different websites are archived at different times, you run the risk that the websites that are referenced on your initial archived website, are from a much later or earlier time). You get an opportunity to get a feeling of how the website was perceived by its

⁴ Niels Brügger, “DIGITAL HISTORIE OG ARKIVERET WEB SOM HISTORISK KILDE”, *DIGITAL HISTORIE OG ARKIVERET WEB SOM HISTORISK KILDE*, 2019.

users. This is not the case with digitized historical sources. Because digitized historical sources often consist of a picture, you, as a historian/user of the digital archive, only get to do you analysis on the written content of that source. This mean that you don't get the same feeling of what context/situation the source was created. With other words you don't get the same amount of meta data.

Lastly, I will just note that one of the other biggest differences between reborn- and digitized historical sources is that reborn digital often are archival multiball times and thus allows the user to see the progression of the archived website, while the digitized source often only are digitized once (this is because the original source behind the digitized version does not change its martial, like websites does)

Bibliography:

Brügger, Niels. "DIGITAL HISTORIE OG ARKIVERET WEB SOM HISTORISK KILDE". *DIGITAL HISTORIE OG ARKIVERET WEB SOM HISTORISK KILDE*, 2019.

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Final Project – Education-level in Aarhus, in the period 1801-1845

Introduction for education in the period 1801-1845.

The typical general understanding of the Danish education system, in the 19'th century, is that it only was for the upper class, and a few lucky ones. But is it true? Was education only for the rich and privileged or was it for the entire population like today? Was education only for men? And did the school law of 1814 influence the society, and thus these questions? In this assignment, we wish to investigate these questions of education-level in the period of 1801-1845 and how it affected the society, children and women⁵.

Background information for the period in interest

Education and school are something that we in Denmark all know very well. But things haven't always been that way and has changed a lot over time. The first law is from 1536 when the reformation took effect⁶. The law imposed the church to make sure that the children of the parish could read, by reading the Bible⁷. But the most important law is from 1814. This law is often recognized as the start of the elementary school in Denmark, because the law secured education for the children, regardless of their background⁸. The school was named "Almueskole". The word "almue" meant people without fortune and education. Children attended school at the age of 7 and ended at the age of 14, when they had been confirmed. Which was an obligation before graduating⁹. But of course, the education system wasn't done that easy, since there was a big difference between the country and the city¹⁰. In the city the children went to school every day, but at the

⁵ Because of limits of the data, we chose to use in this examination of the education level. This research report will be focused on Aarhus (a big city in Denmark, at the time) in the time period of 1801 to 1845.

⁶ "Skoler og undervisning før 1814," Danmarks Nationalleksikon, last visit, 22.maj, https://danmarkshistorien.lex.dk/Skoler_og_undervisning_f%C3%B8r_1814.

⁷ Danmarks Nationalleksikon, "Skoler og undervisning før 1814."

⁸ "Skoler og undervisning efter 1814," Danmarks Nationalleksikon, last visit 22. Maj, https://danmarkshistorien.lex.dk/Skole_og_undervisning_efter_1814.

⁹ Danmarks Nationalleksikon, "Skoler og undervisning efter 1814".

¹⁰ Because Aarhus is a city, we will mainly have our focus on the aspects of education at the time that had an impact on city-schools.

countryside the children were only in school 3 days a week¹¹. Children had chores at home, and they were an important part of labour. A child was often sent out to work before the confirmation. If children did not attend in school, it would cost a penalty, but the work of the children was often worth more than the fine¹².

There were many reasons why the law wasn't implemented quickly. For instance, there wasn't enough teachers, and the seminars could not educate enough teachers in time. So, the church had to educate more teachers along with new priests¹³. In the school the children had to be taught 4 subjects: reading, writing, arithmetic and religion. They were also to learn history about their country and to be loyal to their king¹⁴. The school was also a way to keep control over the citizens. It was strict and violence was normal, even though slapping had been illegal since the school law of 1814¹⁵. The new law did also affect the position of women in society. There were several small private schools in the cities, and these were typically driven by women without education¹⁶. This was because women did not have access to education back then, hence it was only permitted for men¹⁷. But this changed. The needs of more teachers grew which gave unmarried women an opportunity to provide for themselves. The lack of teachers gave women access to the education, not just as teachers, but to education in general¹⁸. In 1899 the school was named its current name

¹¹ Bjerre, Loa, Trine Hyllsted og Jens Aage Poulsen. "Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. klasse", Nationalt Videncenter for Historie- og Kulturarvsformidling, (2014):

https://historielab.dk/wp-content/uploads/2024/02/Hvad-skal-vi-med-skolen-7-9-klasse_100dpi.pdf.

¹² Bjerre, Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. Klasse," 14.

¹³ Danmarks Nationalleksikon, "Skoler og undervisning efter 1814".

¹⁴ Danmarks Nationalleksikon, "Skoler og undervisning efter 1814".

¹⁵ Bjerre, Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. Klasse," 24.

¹⁶ "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne", Skolemonitor, last visit 22.maj: <https://skolemonitor.dk/nyheder/art8201231/S%C3%A5dan-k%C3%A6mpede-kvinderne-sig-ind-i-klassev%C3%A6relserne>.

¹⁷ Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

¹⁸ Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

“Folkeskole”¹⁹. There was a shift. The school was now in name for everyone regardless of their status, rank and background. Education for the entire population didn’t come over the night. We know it took many years to implement the law, because of the economic situation, lack of teachers and it was an entire new system²⁰, so we don’t expect to see a big difference in the first years. Therefore our hypothesis is that we will see a development in work and education through the time-period, maybe not a big one, because it took time, but a development justified by the big changes in the society.

Methods

We started with deciding that we wanted to research the level of education in 1800’s (more specifically from 1801-1860) after having browsed through the provided DAM Seed questions. We also chose this topic because, in the “DAM 2026 SEED EXAM QUESTIONS” document, there was also provided 4 relevant datasets²¹. After having decided on a question/topic, we loaded the 4 relevant datasets into OpenRefine so that we may combine- and clean them for later analysis in RStudio. Since there isn’t a category of education-level, we decided that we would research that types of jobs, that the peoples of the period had as a sign of their education-level.

And after having looked all the dataset through, we estimated that we would have more than enough data, if we used only the datasets from 1801 and 1834, and to lower the size of the dataset that we had to work with, we cut the 1860- and 1845 dataset from our analysis and changed our topic to be focused in 1801-1834.

After having done this, we began cleaning the “erhverv” column, since it was filled with many different values (that however was different variation, of the same general types of occupations). Therefore we decided on clustering together occupation under 13 different

¹⁹ “1800-tallet: Folkeskolen bliver født”, DR, last visit 22.maj: <https://www.dr.dk/ligetil/1800-tallet-folkeskolen-bliver-foedt>.

²⁰ “Uddannelse i 1800-tallet“, Gyldne dage i Præstø, last visit 22.maj, <https://www.gyldnedage.dk/uddannelse-i-1800-tallet/>.

²¹ Aarhus City Archives. (2021). Aarhus City Archives Online Datasets (version 2ec5655). Censuses 1801, 1834 and 1845

categories²² that we meant could represent the different kinds of occupations in the 1800's. To this it should be said that even though we tried to categories all different occupations under the appropriate category, it wasn't all the different variations that Open Refine could/would cluster together. Because of this we afterwards looked through the remaining variations, to make sure that we did not leave out significant data that would have changed the outcome of our analysis. After having done this, we chose to delete the remaining variations since they wouldn't have changed the overall outcome of our analysis and by doing so, we got a cleaner dataset.²³ It should also be said that in the cases where people had multicable occupations, we sorted them under after the occupation that we estimate, required the highest level of education. We then changed the names of the year column to "År" instead of "Ft" and the gender column to "Køn" instead of "Koen", since we wanted an easily readable and understandable dataset and analysis.

At this point, we concluded that we had to reconsider our choice of having cut away the two other datasets²⁴, because our background research showed that it was especially in the later parts of the 1800's that the changes in education showed. We there for decided that we would include the Aarhus-census-dataset from 1845 and thus change our timeframe to 1801-1845²⁵. This meant that we took the "new" dataset and added it to our already processed data in OpenRefine, where we processed it, in the same way that we had done with the other two²⁶. Before we could clean and combine the two datasets, we joined

²² Bonde (all people that supported themselves- and others by means of a farm), Bistandsmodtager (all that got support from the society), Tjenestefolk (all that served someone else), Soldat (soldiers), "lever af sine midler" (alle that lived on either pension or their wealth), Matros (all that lived on the sea), Lærling (all that was in an apprenticeship), Lærer (all teachers), Håndværker (all that produced something with their hands), Familie (all that was supported by their family), Embedsmand (all that had a position in the public administration), Elev (students and pupils) and Academics (all that have taken a higher education).

²³ Because we chose to group occupations together and chose to leave out the remaining the following analysis and diagrams should be seen in a generalising tone since the data isn't precise.

²⁴ That is: The Aarhus-census-datasets from, respectfully 1845 and 1860

²⁵ The reason why we didn't also include the dataset from 1860, was that it was structured in a different way than the other three, and we judge that we would have enough data, by using only the three before mentioned datasets

²⁶ That is: Categorising the occupations under the previously mentioned categories

the English-named columns of the Aarhus-census-1845 dataset with their Danish-named counterparts in our already processed dataset.

After having clean the data in OpenRefine, we loaded the data into RStudio, where we first made RStudio count the frequency of each of the different categories and then sorting it under the associated year and gender. Then we segmented the different occupations, be filtering them one at a time so we could see the amount of people of each gender that fulfilled a given profession. Next, we used ggplot to make diagrams of the spread of occupations. Both in the complete dataset and in each of the researched years. Then, we made illustrations that illustrate the distribution of occupations amongst women and men.

Though our research we also found out that there was an increase in merchant. And because we had sorted them under the “embedsmænd” category, we went back and made a new dataset in OpenRefine, were we only focused on them, and therefor cut out all other occupations so that we were only left with the merchants. After this we loaded the new “Uddannelse_HandelsFolk_1801_1845”-dataset into RStudio where we used the same method as the before mentioned analysis.

Doing our coding, when we ran into problems in the code that we wrote, we used Chatgpt as a sparring partner in fixing the problems in our code, that we couldn’t ourselves diagnose.

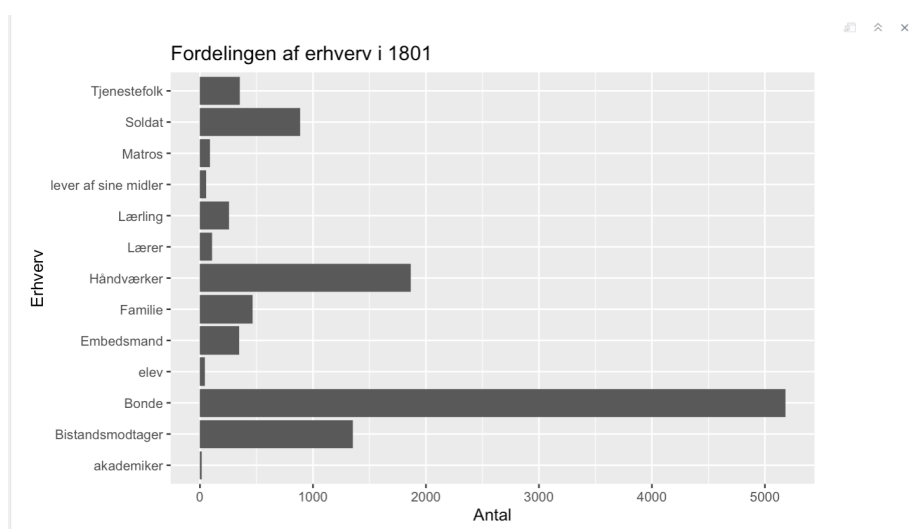
Finding articles

First, we just made a simple google research to find some articles. we were critical against the sides and only used articles and information from websites that was made from museums, archives, lexicons, historians, or some other professional creator, for instance the national museum or The Workers Museum (Arbejdermusset). Many of these is authorized for learning and teaching. We also tried to use website from “Det Kongelige Bibliotek” (The Royal Archive) but didn’t find what we were searching for, because the archive wasn’t in possession of the wanted material. We did occasionally use the chatbot to help us find articles, by describing what information we needed, and afterwards we did find them on google manually, to make sure that they were not made by AI. In the process, we did ask ourselves why these articles were made, who where they created by and for what. But since

most of the articles are from a museum, we concluded that they were made for general information and use for, for example school projects and teaching. Some of the websites given by the chatbot does not have an author connected to the articles, and it is not from any well-known sites, but we concluded the information from the sites corresponds with our own general knowledge about the subject. It is important to be aware of articles with unknown creator. First of all because it is unknown where the author has the information from, and what the purpose of the articles might be.

Our Findings

Graph over profession from 1801.

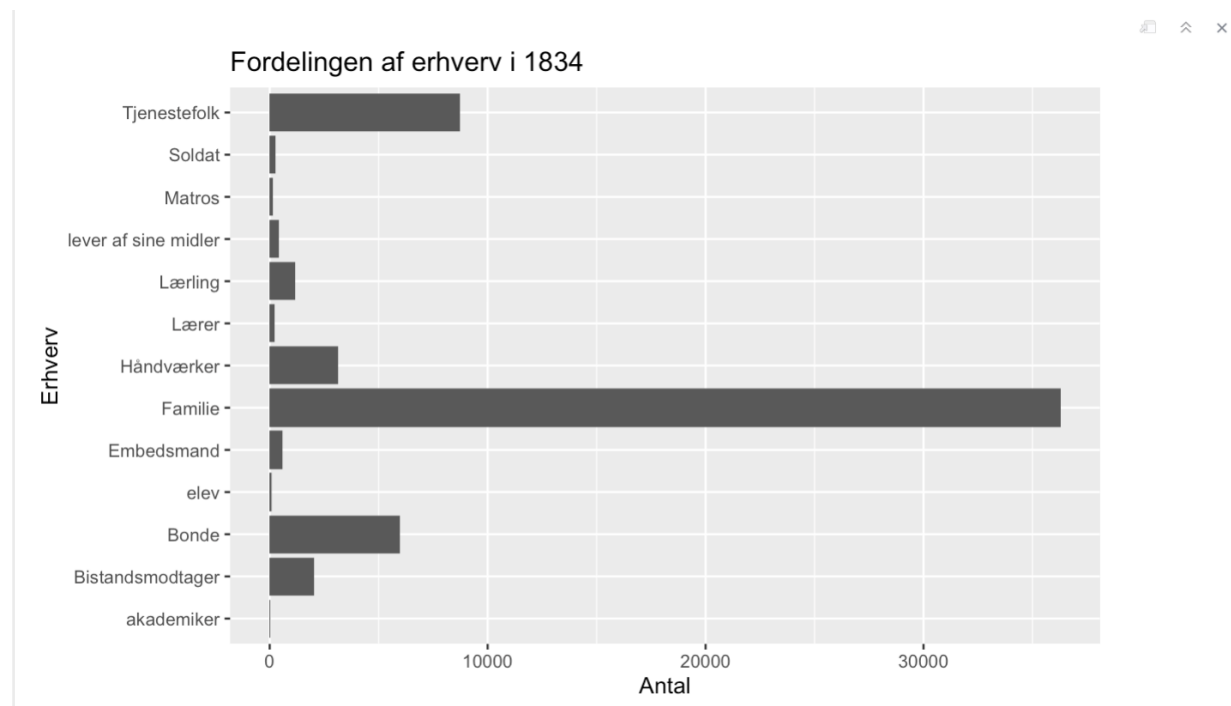


As we can see on this graph. There was a very large group of farmers, which was to be expected. It was the biggest group, and the profession has been indispensable for the population. There was a small group of craftsmen, but it was nearly 2000. It is interesting to see the development of this group. This data is from 1801, so it gives an impression of the condition of society before the law from 1814 about school. So, it's a nice basis of comparison. With this in mind, we shall remember the increase in population. In 1801 there was around 929.000 people in Denmark, but in 1834 that number had grown to 1.231.000 people, in 1840 it was 1.289.000, and in 1850 the population had reach 1.415.000 people²⁷,

²⁷ "Danmarks befolkningsudvikling 1769-2021", Danmarkshistorien, last visit 22.maj:
https://danmarkshistorien.lex.dk/Danmarks_befolkningsudvikling_1769-2021.

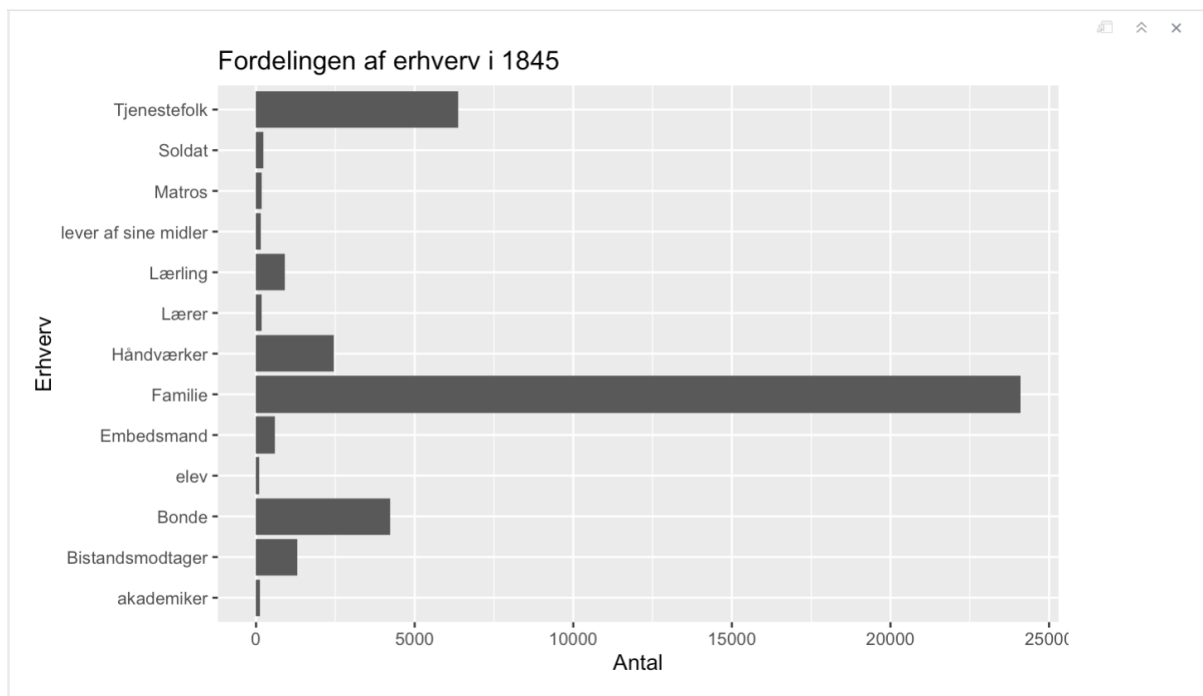
which is interesting since we know many children died during their childhood. We focus on Aarhus, so the population size is smaller, but we can anticipate an increase in the data set from 1801 to 1845.

Graph over profession from 1834.



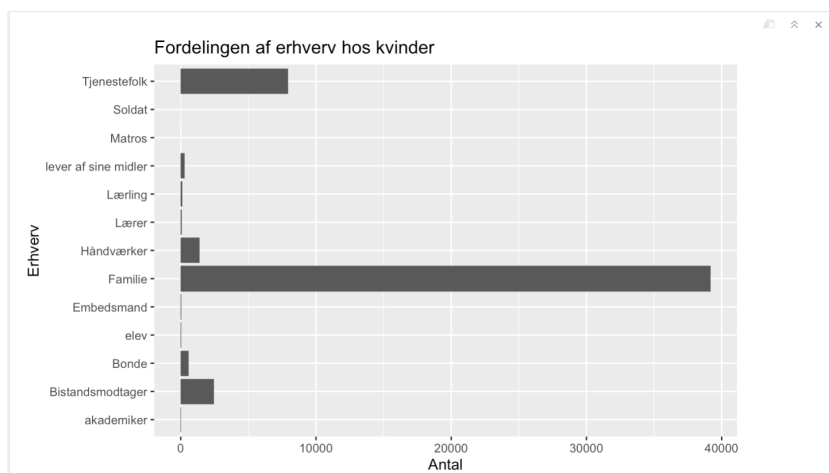
The number of farmers did not change. But the group over craftsmen has grown and there is also the group of “lærlinge” who are to be craftsmen. So, the group of people who are educated has become a little larger. This graph confirms how important the family was in this period. Over 30000 people were dependent on their family to survive. Another important work was as a servant which could be as a maid for the upper middle class. Only a small group did receive money from the public, it was under 2000 people in Aarhus. Only the poorest people could get financial help. It seems that the group of people who gets financial help has grown a bit from the year 1801 to 1834, but it’s hard to read off. Another important thing is that the entire graph is now bigger. Before was it 1000, now it’s 10000. It is because we now have more data. But it’s important to be critical.

Graph over profession from 1845.



It is important to be aware that the number is no longer 10.000. but now 5000. The graph has the same tendency as the one from 1834. The biggest group is still family. The group of servants is still a big part of the labour market since almost 10.000 depended on the occupation. Farmers and craftsmen were also still a major part of the labour market. It's likely farmers who changes business due to industrialisation. The labour market was simpler, and people did not have many career options.

Profession of women.



Most women depended on support from their family. But the group of servants is also decisive. If we compare this graph with the one from 1834, we will see that the number of servants is almost the same, so it's not unreasonable to presume that the numbers from 1834 and 1845 covers the same from this graph of women, and most servants were women, not all, but mostly.

Graph over work for craftsmen, men and women.

A tibble: 6 × 4

År <dbl>	Erhverv <chr>	køn <chr>	n <int>
1801	Håndværker	mand	1477
1801	Håndværker	kvinde	387
1834	Håndværker	mand	2660
1834	Håndværker	kvinde	477
1845	Håndværker	mand	1930
1845	Håndværker	kvinde	511

6 rows

There was, as we saw, a group of craftsmen. The group did contain both men and women. Over the years the group became bigger, maybe after the financial crisis that can have made an impact which forced more people on the labour market. But it wasn't only men on the market, it was also women. When we look at the years and the numbers it's easy to conclude it wasn't a big increase, but more women had attended on the market. It can have been as seamstresses, dressmakers or cleaner in a laundry. Some work did require some kind of education, which made women more independent because they earned their own money. But for men it was the opposite. There is a huge fall. But why? We didn't expect that to happen.

Graph over "elev" (student).

A tibble: 6 × 4

År <dbl>	Erhverv <chr>	køn <chr>	n <int>
1801	elev	mand	36
1801	elev	kvinde	9
1834	elev	mand	47
1834	elev	kvinde	19
1845	elev	mand	99
1845	elev	kvinde	3

6 rows

It was a very small part of children who attended school, both before and after the new law. It's remarkable that the number increases from 1801 to 1834 but falls after for female students. It went from 9 to 19 and then falls to 3. But at the same time the number of male students increased the entire period. Why could that have happened?

Graph over "lærlinge".

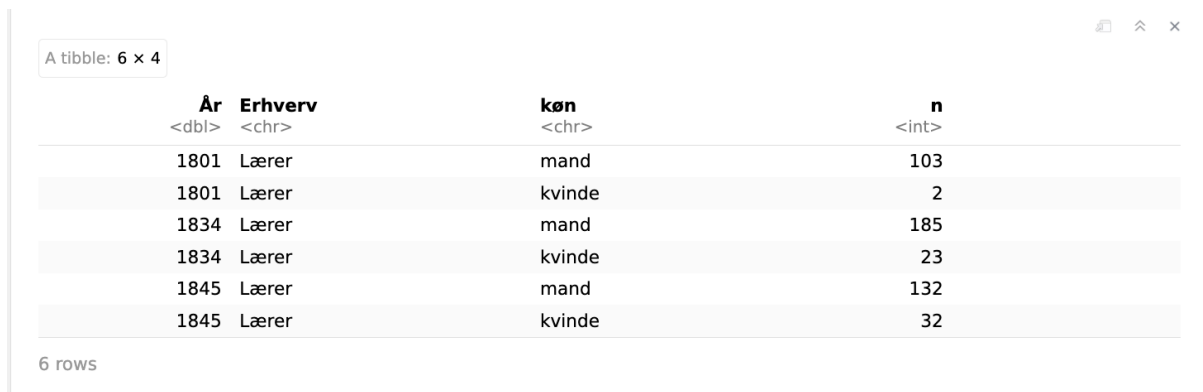
A tibble: 6 × 4

År <dbl>	Erhverv <chr>	køn <chr>	n <int>
1801	Lærling	mand	234
1801	Lærling	kvinde	20
1834	Lærling	mand	1137
1834	Lærling	kvinde	47
1845	Lærling	mand	867
1845	Lærling	kvinde	39

6 rows

This graph shows the same tendency as the one for "elev". The number is increasing the first period but falls in the second period. Why? Financial reasons? Economy? Statistic? This will be addressed in the discussion

Graph over teachers.



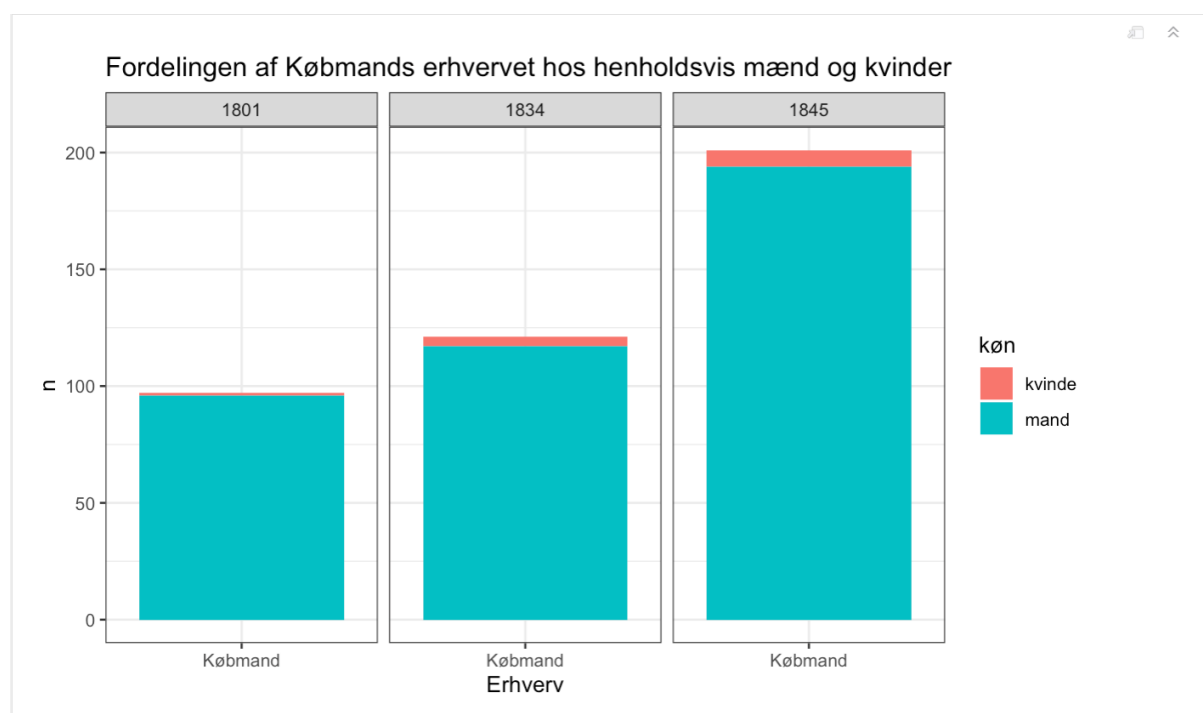
A tibble: 6 x 4

År <dbl>	Erhverv <chr>	køn <chr>	n <int>
1801	Lærer	mand	103
1801	Lærer	kvinde	2
1834	Lærer	mand	185
1834	Lærer	kvinde	23
1845	Lærer	mand	132
1845	Lærer	kvinde	32

6 rows

This graph is very distinctive. The number of female teachers exploded in these years. Of course, in number it's not that big, but in percentage it was a big increase. Women did now have access in this part of the labour market. But not the entire school system was available for women, it was mainly small private schools. The progress is probably also because there was a huge need of more schools and teachers after the new law. Of course, that was related to the public schools, but we know many parents was dissatisfied with the public school system and put their children into private schools instead, because the private sector contained more subjects and the children went to school for more years. This development turned out to benefit women, but it's still important not to draw this conclusion too fast, because there is a fall in the number of male teachers. The fall was bigger, than the increase of female teachers.

Discussion.



As shown on the model above, the occupation “købmand” clearly did not grow much over the span of 33 years from 1801 - 1834. This has something to do with the time period. In this period happened the Napoleonic wars, the Bombardment of Copenhagen, state bankruptcy in 1813 and the loss of Norway in 1814. The biggest export was corn to Norway, but in 1814 England imposed a tariff, which made it hard to be a merchant. All these events caused many merchants to go bankrupt in the following years, and their job depended on keeping turnover small and expenses low, which may be a cause that this “erhverv” did not grow significantly during these years. It also tells a story about society in a broader perspective. Being a merchant was just one of few occupations at the time, but all social classes and occupations were affected by the events in the early 1800 hundreds. The model above about merchant occupation is therefore also a clear example and is a proof of a low economy throughout society, which is the reason there was no money to fund education.

As displayed the merchant almost doubled in the period 1801-1845. This happened for several reasons. One of the main reasons was that the merchants for a safety measure never put all their funds in a single business, meaning they would invest in two or more completely different things, to ensure they didn't go bankrupt even if one of their businesses failed. Along with this the merchants used their influence to build the harbour in

Aarhus larger to make it more attractive for foreign trade during the 1840's. England also ultimately lifted their tariff on corn in 1846 and made a direct steamship connection in 1848. The new connection made sure that England became the largest trading partner to Denmark during the war with Germany in 1848-1850 which was important for the city's merchants.²⁸ Therefore, during this period, the education system became better and larger, because the society and economy grew as a whole.

Was education for the upper class? Was it everyone? How did the changes in the society affect the life of the population after the law from 1814? That is what we wish to discuss based on the data and background knowledge. We know that all children by law were to go to school after 1814. But did they come in school after all? We know from the data that there was only a small group of "elev" in 1834, and a little group of "lærlinge". With that knowledge it's not unreasonable to suppose that not all children came to school, but why did they not? Was it because of economic or practical reasons? The children in Aarhus were to go to school every day, but could it be true that there was such a small group of children in school in Aarhus? The first years after the new law it was hard to implement the new law because there was a lack of teachers, buildings and money after the state bankruptcy²⁹. We saw an increase of teachers in the years between 1801 and 1845, but why, if there wasn't that many children in school? That is a remarkable development. What was the reason? In the start there could be a long distance from the school and the children, which might have prevented some children from going to school, we assume this because it was decided that a child should not have more than 1,8 km³⁰ to school and why introduce this law without cause.

²⁸ "Ugens Aarhushistorie: Erhverv Aarhus 160 år – Handelsstanden drev byens udvikling frem. Aarhus stadsarkiv." Updated january 2024: <https://stadsarkiv.aarhus.dk/nyheder/2022/ugens-aarhushistorie-erhverv-aarhus-160-aar-handelsstanden-drev-byens-udvikling-frem>

²⁹ "Uddannelse i 1800-tallet", Gyldne dage i Præstø, last visit 22.maj, <https://www.gyldnedage.dk/uddannelse-i-1800-tallet/>.

³⁰ "Undervisningspligt", Skolehistorie, last visit 22.maj: <https://skolehistorie.au.dk/temaer/skolepolitik-og-skolefolk/undervisningspligt>.

Many families depended on the work of children, who was an important part of the household³¹. Some children were sent out to work before the confirmation and before they had finished school. Often, because the family couldn't do without the pay³². In some cases, the parents might have chosen to keep the child at home to work instead, even though it would cost a penalty. But, if the work was more valuable, that would be the consequence³³. This might be one of the factors why the group of children were so small, but there were other groups involved. We know from articles that some parents were dissatisfied with the public school and sent their children to private schools, instead³⁴. That must have been the upper class, who did that. Because they were the part of the society, who could afford that. Not only could these families, do without their children's work and help to the household, but they could afford to spend money on the child's education. That was not something the average family could afford. They depended on the child as a part of the economy³⁵.

Between 1788 and 1800, the serfdom(stavnsbåndet) was abolished. Serfdom meant that farmers (bonde) were bound to their estates, meaning they had to remain on the estate where they were born. This meant, that their freedom of movement – and employment opportunities – were very limited. With the abolition of serfdom, farmers were now free to choose where they wanted to work or seek employment. This happened alongside a significant population growth³⁶. In the 1800 hundreds, because of the population growth, there was a significant surplus of labour in the countryside, where many younger children

³¹ "Børneliv", Nationalmuseet, last visit 22.maj:

³² Bjerre, Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. Klasse," 14.

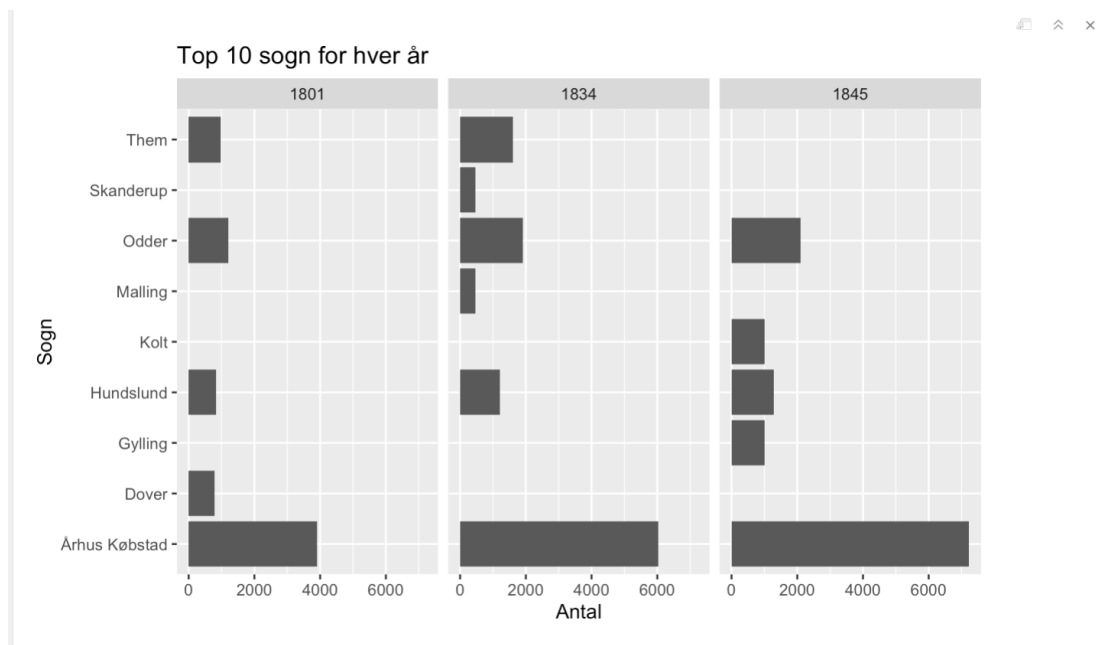
³³ Bjerre, Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. Klasse," 14.

³⁴ DR, "1800-tallet: Folkeskolen bliver født".

³⁵ Bjerre, Hvad skal vi med folkeskolen? - om samfund og skole før, nu og i fremtiden 7.-9. Klasse," 14.

³⁶ "Ophævelse af stavnsbåndet 1788-1800". Danmarks Nationalleksikon. Updated March 2025:https://danmarkshistorien.lex.dk/Oph%C3%A6velse_af_stavnsb%C3%A5ndet_1788-1800

had to move to larger cities to find work and support themselves through other professions professions(erhverv).³⁷



The graph above illustrates the population growth in the years 1801-1834-1845. This is used as proof that there was a great growth in the smaller cities and especially in Aarhus. Aarhus almost grows with double the amount of people, which is an example of that smaller cities grew to such and extend, that many people had to move to the bigger cities to find a way to support themselves.

At the same time, industrialization was taking place. In Denmark, it began around the 1830's, creating a greater need for labour in the cities, which also made it an obvious opportunity for farmers to seek better living conditions.³⁸ As a result, the graphs (fordeling af erhverv i 1834 /1845 - Seen under "methods") show a decline in the number of farmers from 1834 to 1845 and there is an increase in other professions. The graph in top of the

³⁷ "Den industrielle revolution". Danmarks teknologi historie. Last visit 22. May: [Den industrielle revolution – Danmarks Teknologihistorie](#)

³⁸ "Industrialiseringen i Danmark – En Historisk Oversigt". Familie Minder. Last visit 22. May: [Industrialiseringen i Danmark – En Historisk Oversigt](#)

discussion (Fordeling af købmænds erhvervet hos henholdsvis mænd og kvinder), shows that some of these people for instance could've chosen the merchant profession. The proportion of the population that has taken another profession corresponds to the decline in the number of farmers, which is consistent with the fact that farmers had moved to the cities to improve their opportunities rather than continuing farming.

School in Aarhus.

Before the new school law in 1814, there were only two options for children in Aarhus. Either they had to go to a private school, which only the wealthier families could afford, or to a charity school (fattigskole). Because of the new school law, of 1814, they had to come up with a new system and therefore was a school directorate, consisting of district magistrate (Stiftamtmand) F. C. J. Güldenkrone and the provost O. A. Secher in charge of putting down a school commission. The commission consisted of the city clerk (Byfogeden), the city councillor (Rådmanden) and five appointed citizens. The commission's proposal was to build one big school that could house 300 students at once. The proposal did come with certain concern. Because as mentioned above there was a lack of teachers during this time and the conclusion was therefore that a single teacher probably could not control 60-70 students during recess, and therefore also could not control 20 students in a classroom. Along with this, the commission also suggested to build secondary schools (realskoler) for the upper-class children to get a preliminary examination in which the commission collected citizen signatures for approval. The third suggestion was to also keep the charity school. The problem with their plan was that this would become very expensive, and after the state bankruptcy in 1813 there was no money to invest in schools, both in Aarhus and the rest of the country. The proposal was therefore put down.

The matter of a new school system was brought up again in 1817 since it was an obligation to act on the school law from 1814. The school directorate therefore came up with a new plan that differed from the school commission's plan. They scrapped the plan of secondary schools since it was too expensive and would instead build two civic schools (borgerskoler), which each could contain 150 students, and they would also keep the charity school. They decided to build two schools instead of one, since some children would have a long way to go to school, and their concern was that it would keep some of the children from attending

class. This proposal was highly criticised by the school commission, because they thought that the upper-class children should not be treated so indifferently, and offered better education. Even though the school directorate faced this criticism they still decided on this plan, and with the approval of the royal chancery, this plan was finally approved in May 1818.³⁹

Women in school.

Women did not have access to education, and definitely not an academic education, which was reserved for men⁴⁰. But why did men leave the work as teachers, as we clearly saw in the before referenced a data? Why did the number of female teachers suddenly grow? Economy reasons, practical reasons, private schools, maybe? After the law was introduced in 1814 it took years to implement the law, one of the reasons was lack of teachers. There was established several private schools⁴¹, which stimulated the need of teachers and maybe that made the society accept female teachers because of the lack of male teachers. Even though there already were a few female teachers for example in private homes⁴². What motivated the women to work? Probably economy since there had been a crisis after the Napoleon-wars and Denmark went state bankrupt. But also, the marriage status. If a woman was unmarried and wasn't provided for by her family, she had to support herself. But not everyone was positive about women taking an education. The actor and author Johanne Louise Heiberg was one of them. She encouraged mothers to keep their girls at home and find a marriage match⁴³. In the 1840s women got access to the education as a teacher, and

³⁹ Unknown Author. "Skænderiet om byens første borgerskoler ." *Aarhus Stiftstidende*. (2014): 44, https://stadsarkiv.aarhus.dk/media/jhnpvclw/aarhuswiki_stadsarkivet_stiften_2014.pdf

⁴⁰ Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

⁴¹ DR, "1800-tallet: Folkeskolen bliver født".

⁴² Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

⁴³ Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

that became important for women even taking an education⁴⁴. So why wasn't there not as much male teachers as before? Maybe even after the critic a tendency has started against teaching as a woman occupation? Maybe some male teachers went into the private sector, or the graph covers the tendency with teachers educated by priests, and when that arrangement was lacking towards an end, there was fewer male teachers. But one thing is for certain, according to our statistic, in the later part of the period in question, the number of male teachers fell. But it is significant that there were not more female teachers. Another reason for the late implement of the law was economy. Copenhagen had been bombed. Denmark was state bankrupt and Norway was lost. But there was still introduced a new law that would turn out to change the society. The state wanted to make sure that the children got a basic education and was loyal to the country and the crown, but the state wasn't willing to invest money in the school system, or were they? They built new schools, but not all teachers were educated, and the teaching was in basic subjects⁴⁵. This made the parents put the children into private schools⁴⁶. But as we see an economic stronger country, the number of teachers grew, but not the number of female students. The number fell, as we concluded in our findings. It hard to tell why. It could be a mistake in the data. It's from the 19th century so maybe there was a mistake when it was written or digitalized and who were responsible for collecting the data. But why is it only girls then, and not boys too? That makes the argument about a mistake less likely. It could be after the tendency where some children were put into private school or educated at home, and maybe they do not appear in the data after that. We can't say for sure, but that is the most plausible cause.

The graphs shows that people changes business as time passed on, and they got more educated. Some went to be "lærlinge" before they were educated as some kind of craftsman (håndværker) and the number of them grew in the first years but fell in the last period. The entire population learned to read, write and compute. So, education changes the society. Remarkably, there was a fall in the groups of male craftsmen and the entire

⁴⁴ Skolemonitor. "Fire nedslag i skolehistorien: Sådan kæmpede kvinderne sig ind i klasseværelserne".

⁴⁵ DR, "1800-tallet: Folkeskolen bliver født".

⁴⁶ DR, "1800-tallet: Folkeskolen bliver født".

group of “lærlinge”. Which came as a surprise, because we did not expect that to happen. We thought there would be an increase. A reason for the fall could be because some craftsmen turned over to trade after the economy became better. But about the “lærlinge”? If there was a shift on the market, and more people turned over to a new profession, there would probably not be many apprenticeships. So, the number might not cover over people who left the apprenticeship, but a fall in how many who got an apprenticeship. We can't tell for sure, but that is the most likely explanation.

Lessons learned

In working with this data analysis assignment, we have learned much. For example, it quickly became clear how important it is to have a good grasp on your background reading before, analysing your data. For example, we had to start over on the data cleaning and analysis a few times, when it became clear that we either had removed or combined data that we needed for our project. An example is that in the first draft of our data cleaning, we had combined the occupation “købmand” with the occupation “embedsmand”. We therefore had to spend time to make a new dataset with more separated occupations and more individual categories. We also learned that OpenRefine lacks a few functions, since it does not contain every aspect relevant for your dataset. The algorithm OpenRefine uses for clustering data could be better. We learned that ChatGPT is a great tool for coding assistance, and we made sure that the articles AI refers to actually exist and that we were able to find them ourselves and confirm the data is reliable. We were also impressed by occupation situation during our time period, there was a greater complexity that we expected, since some of the occupations would fall in numbers, though the reason for this could be a lack of data.

Conclusion

Overall, we concluded that the hypotheses that we started with held true. This can for example, be seen in the steady increase in the amount of “elev” from 1801 to 1845, which confirms our background-reading's claim, that there in the 1814 came a law that forced children into public primary schools. We also conclude that, though our research of the provided census-datasets from the 1800's, we that education in the 1800's – in the start – was mainly withheld for the upper class of society, but also that slowly, throughout the

century, this started to change. We also conclude that, through our research of the provided census-datasets from the 1800's, that education in the 1800's – in the start – was mainly withheld for the upper class of society, but also that slowly, throughout the century, this started to change. We concluded, after having done data analysis, that it showed a steady increase in the amount of "elev", "lærling" and "Købmand" from 1801 to 1845. This, compared to the drop in "bonde", shows a shift to occupations requiring higher levels of education.

In Aarhus the school options for children were very limited before the school law of 1814, but the new law still gave major problems the city. There were two different plans of how the new school system should be and therefore disagreements between the school directorate and the school commission. Ultimately it was the school directorate's plan that were put to work, and it was decided that there should be two civic schools instead of one, alongside with the charity school and no secondary schools for the upper-class children. We conclude the main problem of this conflict was grounded in financial crisis, but the final plan was approved by the royal chancery in 1818. Since it was the school directorates plan that was put to work, we also conclude that the education in Aarhus was for all children, and not just the privileged or upper-class.

Admission to education changes the terms for women. Before, education had only been an opportunity for men, and a woman could only perpetrate in the private system, but in the 1840s women got access to the exam in the public system, even though many criticized it and reactionary people wanted women to stay in the home. In conclusion women did now have better opportunity to support themselves, which changed the dynamics between the genders.

Lastly, we conclude that education availability increased in the period but compared to today there were still limitations on higher levels of education, in both who got the opportunity and who had the means necessary.

Author contributions

Conceptualization: All Authors

Methodology: Kristian Wolf Krogh, Amanda Rundstrøm Hjortgaard Nielsen, Emil Jensen

Digital data wrangling, analysis and visualisation: Kristian Wolf Krogh

Github-responsible role: Kristian Wolf Krogh

Writing

- Historical context: Emil Jensen, Amanda Rundstrøm Hjortgaard Nielsen
- Results: Emil Jensen, Amanda Rundstrøm Hjortgaard Nielsen
- Discussion: Emil Jensen, Amanda Rundstrøm Hjortgaard Nielsen
-

Writing - critical review and overall editing for uniform voice: Emil Jensen, Amanda Rundstrøm Hjortgaard Nielsen, Kristian Wolf Krogh

Table 1 – Software metadata

Software metadata description	Please fill in this column
<i>Permanent link to Github repository where you put your script(s), R project, and data</i>	GITHUB-Link
<i>Software License</i>	R: Licens: GNU General Public License v3 (open source)

	Rstudio: Licens: GNU General Public License v3 (open source) OpenRefine License: BSD3-Clause "New" or "Revised" License
<i>Data License</i>	<i>The data we have used is licensed under:Public Domain Dedication and License (PDDL)</i> https://github.com/aarhusstadsarkiv/datasets/blob/master/LICENSE Our processed data are licenced under:
<i>Software versions, Installation requirements & dependencies for software not used in class</i>	<i>You need to have RStudio(version: 2026.04.0+526) installed</i> <i>You need to have OpenRefine(version: 3.10.0) installed</i> <i>You need to install the following packages installed into Rstudio:</i> - Tidyverse - Ggplot - dplyr
<i>If available Link to software documentation for special software (only relevant if you go outside the scope of class)</i>	We have not used any software that goes outside the scope of class
<i>Support email for questions</i>	Au791343@uni.au.dk

Table 2 – Data metadata (use the template below or create your own metadata table)

Metadata description	Please fill in this column
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<i>census-1801-normalized.csv</i>	Census data on people from Aarhus in the year of 1801, collected by the Danish government in 1801 with the 17 collumns: ft, sogn, amt, id, loknr, lokalitet, bygning, famnr, fnavn, enavn, koen, famstad, alder, civilstand, giftnr, erhverv, notater
<i>census-1834-normalized.csv</i>	Census data on people from Aarhus in the year of 1834, collected by the Danish government in 1834 with the 16 collumns: ft, sogn, amt, id, loknr, lokalitet, husnr, bygning, famnr, fnavn, enavn, koen, alder, civilstand, erhverv, notater
<i>census-1845-normalized.csv</i>	Census data on people from Aarhus in the year of 1845, collected by the Danish government in 1845 with the 17 collumns: county, parish, id, location_number, location, building, house_number, Family_number, lastnames, gender, firstnames, age, marital_status, occupation, notes, county_of_birth, parish_of_birth

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All datasets are available on Github in CSV format, authored by Aarhus Stadsarkiv, og Claus Juhl. "Datasets/Censuses/ Aarhusstadsarkiv/Datasets". GitHub. Seen 22. may 2026.

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- Can also be found in our GitHub repository, under the folder:"Raw-Data"

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Chatgpt (free Web version)

- We have used the free web-browser version of Chatgpt, to fix mistakes in our code and as a sparring partner on the capabilities of OpenRefine and RStudio. We have also used it as a sparring partner in the way that we asked to come with suggestions to possible editing

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Deklaration for anvendt Generativ Kunstig Intelligens (GAI)

– gældende version fra 4. juni 2025

Brug af GAI i projekter

I henhold til de [nye regler og anbefalinger på Aarhus Universitet](#) omkring brug af værktøjer baseret på Generative Artificial Intelligence (GAI) skal du vedlægge en deklARATION som bilag til opgavebesvarelser, der angiver, om du har anvendt sådanne værktøjer.

Bemærk dog følgende: Hvis du ikke har brugt GAI-værktøjer, skal du *ikke* vedlægge en deklARATION, ligesom der *ikke* forventes en deklARATION ved stedprøver.

Hvis du *har* brugt GAI-værktøjer, skal du i deklARATIONEN angive, *hvilke* værktøjer du har brugt, og *hvordan* du har brugt dem.

Ved at deklARERE din brug af GAI-værktøjer sikrer du, at der ikke opstår udfordringer i forhold til reglerne om eksamenssnyd.

Du vil ikke blive bedømt på din måde at bruge GAI-værktøjer på. Din opgavebesvarelse vil som altid kun blive bedømt ud fra læringsmålene i kursusbeskrivelsen eller studieordningen. Formålet med deklARATIONEN er udelukkende, at du godtgør, at du har forholdt dig til, om du har anvendt GAI-værktøjerne på en fornuftig måde.

Du skal dog være opmærksom på, at der i forbindelse med visse kurser, hvor GAI er integreret i fagligheden, er en forventning om, at en refleksion over anvendelsen af GAI-værktøjer indgår som en del af et metodeafsnit. Du skal huske at tjekke, hvorvidt det er tilfældet, ved at spørge din vejleder eller din underviser.

Opmærksomhedspunkter:

- Husk altid at undersøge gældende regler og retningslinjer for brug af GAI på studerende.au.dk.

- Du kan også opsøge information på din studieportal.
- Vær opmærksom på ikke at fodre GAI-værktøjer med personfølsomme eller andre fortrolige oplysninger.
- Hvis du har brugt GAI-værktøjer til at genere tekst, skal du deklarere, hvordan du har brugt værktøjerne. Husk at følge [reglerne](#) for korrekt parafrasering.
- Hvis du har citeret direkte fra tekst, der er genereret af GAI, skal du angive det som citat med en reference til det GAI-værktøj, der er benyttet – se vejledning på [AU Library](#).

Sådan deklarerer du brugen af GAI-værktøjer

I din deklaration af brugen af GAI-værktøjer, skal du som minimum:

1. tilkendegive, at du har brugt GAI,
2. angive, hvilken teknologi du har brugt,
3. beskrive, hvordan information er blevet genereret, beskrive de anvendte input samt forklare, hvordan du brugte outputtet i opgavebesvarelsen.

Du skal bruge følgende skema, når du laver din deklaration.

Deklaration for anvendelse af GAI-værktøjer

- | |
|--|
| <p>☐ Jeg/vi har benyttet generativ kunstig intelligens til udfærdigelse af dette projekt (sæt kryds). Oplist, hvilke GAI-værktøjer der er benyttet (husk version):</p> |
|--|

Jeg/vi har brugt GAI-værktøjer på følgende vis) – *en liste over mulige anvendelser er vedlagt til inspiration*

EKSEMPLER PÅ ANVENDELSE
Til sparring ifm. formulering af problemstilling
Til sparring ifm. valg af teori og metode
Til feedback på egen tekst X
Til at få alternative formuleringer
Til at komme i gang med at skrive
Til at forstå et emne bedre X
Som en hjælp i læseprocessen X
GAI er brugt som aktuelt emne i projektet
Til at finde videnshuller
Som hjælp til at starte min/vores søgning
Til programmeringsopgaver X
Til dataanalyse X
Til fremstilling af figurer X
Anden anvendelse (angiv hvilke(n)):

For hvert relevant område forklares, hvordan GAI er benyttet (se forklaringer af de forskellige anvendelsesmuligheder på [Studypedia](#)). Beskriv f.eks. kortfattet, hvordan informationen blev genereret, og forklar, hvordan outputtet er anvendt i din/jeres opgave.

Til feedback på egen tekst:

- Vi fik GAI til at give os feedback på vores tekst, ved hjælp af det prompt som Adela havde givet os

Til at forstå et emne bedre

- Vi har brugt GAI til at forstå og refine bedre

Som en hjælp i læseprocessen

- Vi har brugt GAI til at hjælpe med at finde artikler og relevant materiale. Derudover brugt til at forklare tekster

Til programmeringsopgaver

- Vi har brugt GAI til sparring samt effektivisering af vores koder. Derudover har vi også fået hjælp når vores koder ikke har virket.

Til dataanalyse

- Vi har brugt GAI om hjælp til refine

Til fremstilling af figurer.

Danish Monarchs

- Jeg brugte AI til at få oversat opgaven til dansk og forstå opgaven, samt jeg fik hjælp til at fremstille graferne. Jeg brugte IKKE AI til at skrive teksterne som graferne er forsynet med. Jeg brugte den som en sparringspartner.